Economic and Agricultural Determinants of Renewable Energy Consumption:

Evidence from a Global Panel, 1995–2020

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Abstract

This paper investigates the relationship between economic development, agricultural conditions, and renewable energy consumption using unbalanced panel data from 122 countries during the period 1995–2020 (N = 3,083 observations). Data sources in this study were compiled from the World Bank and Mendeley Data (v1). Changing the perspective from previous studies, this paper focuses on analyzing the multidimensional links between economic dynamics, the agricultural context, and the renewable energy transition process. The observation unit is country-year. Using pooled OLS estimation methods, two-way fixed effects (FE), and first-order difference (FD), I found three main results. First, the relationship between GDP per capita and renewable energy consumption is nonlinear and U-shaped with an estimated inflection point of approximately US$8,860–12,000/person. Second, per capita CO_2 emissions are negative and strongly correlated with renewable energy consumption in all models (FD: −0.43 pp/ton, $p < 0.001$), reflecting dependence on fossil fuels. Third, agricultural value-added positively predicts domestic renewable energy growth according to the FD model ($\beta = +0.122$, $p < 0.001$), consistent with the biomass energy channel, but this effect is strongly reversed in developed countries (FE: −1.228, $p < 0.001$), where agricultural growth reflects fossil fuel adoption rather than biomass. These findings suggest that agricultural development policies in low-income countries may serve as an additional benefit to the renewable energy transition.

I. INTRODUCTION

The transition toward renewable energy has accelerated since the 1990s, driven by environmental pressures, technological advancements, and shifting economic incentives. However, the pace of this transition varies significantly across countries. Understanding the economic and structural conditions that facilitate or impede renewable energy adoption is therefore central to designing effective climate and energy policies.

A vast empirical literature links macroeconomic development to energy mix transitions. Cross-country evidence demonstrates that GDP growth, trade openness, and urbanization are key drivers of renewable energy expansion (Aguirre & Ibikunle, 2014; Marques et al., 2010). More recently, analyzing a panel of 128 countries from 1990 to 2020, Ahmad and Majeed (2024) find that economic openness and urban population shares positively predict renewable energy production, whereas fossil-fuel electricity generation exerts a crowding-out effect. Wang et al.

(2020) also document a non-linear relationship between income and renewable energy consumption. A common limitation of this strand of research, however, is its narrow focus on economic factors, largely overlooking agriculture and land-use dynamics; even though biomass energy constitutes a substantial share of renewable energy consumption in developing economies (IEA, 2022).

This study contributes to the literature in three ways. First, I directly integrate agricultural variables alongside standard economic indicators to examine whether agricultural activity serves as a channel for renewable energy adoption. Second, by employing two-way fixed effects and first-difference models, I exploit within-country variation over time, thereby providing cleaner identification of the agricultural channel compared to the cross-sectional correlations documented in prior work. Third, I uncover structural heterogeneity based on development status: the agricultural channel operates in opposite directions for developing and developed nations.

II. RESEARCH QUESTION & REPORT STRUCTURE

Three research questions guide this analysis:

1. Which economic and agricultural factors are associated with global renewable energy consumption over time?
2. Do these relationships systematically differ between developed and developing countries?
3. Does agricultural value-added capture a within-country, year-on-year transition toward biomass energy?

The remainder of this paper is organized as follows:

- Section III describes the data and presents descriptive evidence.
- Section IV outlines the empirical strategy and regression specifications.
- Section V reports and interprets the main results.
- Section VI discusses heterogeneity analysis and robustness checks.
- Section VII concludes.

III. DATA

A. Sources and Sample

The dataset used in this study is drawn from the publicly available dataset of Wang (2025), published on Mendeley Data (DOI: 10.17632/k34hgzgz6h.1), which was compiled to support research on the relationship between climate change and energy security. The original dataset integrates information from two primary sources: economic and agricultural variables are extracted from the World Bank's World Development Indicators (WDI), while energy-related indicators are supplemented with data from Our World in Data (OWID). This study adopts the variable structure and definitions from that dataset directly, with several additional modifications including the removal of anomalous observations, construction of derived variables, and income-group classification.

The final estimation sample comprises 122 countries over the period 1995–2020, forming an unbalanced panel with the number of observed years (T) ranging from 9 to 26, yielding a total of 3,083 country-year observations. Six countries are excluded due to systematic data gaps across core variables: Djibouti, Montenegro, St. Lucia, Myanmar, St. Kitts and Nevis, and Equatorial Guinea. Income classification follows World Bank thresholds: 41 countries are designated high-income (GDP per capita ≥ $12,476, constant 2015 USD) and 81 countries are classified as low- or middle-income.

B. Variables

Dependent variable. The outcome of interest is renewable energy consumption as a share of total final energy consumption (%), denoted renew. The sample mean is 21.4 percentage points (pp) with a standard deviation of 20.6 pp; the median is substantially lower at 15.3 pp, indicating a right-skewed distribution. Two observations with negative recorded values (data anomalies) are set to missing. A log-transformed version, ln(renew), is used in robustness checks.

Economic regressors. The key economic variable is the natural logarithm of GDP per capita (measured in constant 2015 USD), along with the square of this variable, included to capture the non-linear relationship between GDP and renewable energy. Additional economic control variables include net FDI inflows as a share of GDP, the urban population ratio, and trade openness (% of GDP). This final variable has 11% missing observations and is included only in robustness check models.

Agricultural and climate variables. Agriculture value added as a share of GDP (WB: NV.AGR.TOTL.ZS) is the main agricultural regressor of interest. Forest area as a share of land area proxies the structural availability of biomass. Average annual temperature (°C) and precipitation (mm/yr) capture climatic conditions relevant to both hydropower and biomass potential. These latter variables change slowly within countries; they are therefore excluded from within-estimators (FE and FD) where they would be collinear with country fixed effects.

Controls. CO_2 emissions per capita (metric tons) serves as a proxy for fossil-fuel intensity of the energy mix. The natural logarithm of total population controls for country size. Both enter all specifications.

Table 1. Descriptive Statistics

Variable	N	Mean	Std. Dev.	Min	p25	Median	p75
Renew. energy consump. (%)	3083	21.43	20.58	0.00	4.16	15.33	33.02
ln GDP per capita	3083	9.02	1.16	6.35	8.21	8.88	9.90
FDI, net inflows (% GDP)	3083	5.49	17.49	-57.61	1.22	2.90	5.78
Urban population (%)	3083	63.50	19.36	17.89	50.67	65.85	78.48
Trade (% GDP)	2841	87.91	49.90	0.03	56.79	78.64	103.24
Forest area (% land)	3083	31.72	23.29	0.00	11.69	30.46	49.75
Agri. land (% land)	3083	37.94	21.60	0.44	19.62	39.93	53.93

Agriculture VA (% GDP)	3083	7.78	7.20	0.03	2.22	5.55	11.16
Temperature (avg. °C)	3083	17.54	8.32	-5.17	10.34	19.88	25.26
Precipitation (mm/yr)	3083	1158.13	900.09	14.01	478.46	874.81	1739.6
CO2 per capita (metric t.)	3083	5.79	5.65	0.14	1.78	4.11	7.85
ln Population	3083	15.85	1.99	11.14	14.78	15.87	17.28

Notes: Sample: 122 countries, 1995–2020. Statistics computed over country-year observations with non-missing values. All monetary variables in constant 2015 USD.

C. Descriptive Evidence

Table 1 presents summary statistics for the main variables. The large standard deviation of *renew* (21.4pp) reflects substantial cross-country heterogeneity in energy portfolios. A variance decomposition shows that approximately 95 percent of total variation in *renew* is between-country, with only 5 percent within-country over time: an important constraint on the identification strategy discussed in Section III.

Figure A.1 documents the distribution and time trend of the dependent variable. The distribution is markedly right-skewed (mean = 21.4%, median = 15.3%), driven by a small group of low-income countries with large traditional biomass sectors. The log transformation substantially reduces this skewness, justifying ln(*renew*) as a robustness check. Across five-year intervals, developing countries consistently report higher median renewable shares than developed countries, reflecting biomass dependence rather than modern renewables. Developed countries show a modest upward trend from 2010 onward, consistent with solar and wind expansion.

Figure A.2 previews the headline result. The scatter of ln(GDP per capita) against *renew* reveals a U-shaped relationship suggests that renewable shares decline through the middle-income range then recover at higher incomes, which motivating the quadratic specification used throughout.

Figure A.3 shows bivariate evidence for the agricultural channel. Forest area and agriculture value added are both positively associated with renewable consumption, with the latter exhibiting the stronger relationship (r = +0.443***). The time trend in panel (c) confirms that low- and middle-income countries maintain persistently higher renewable shares throughout the sample period, consistent with biomass dependence rather than modern renewables.

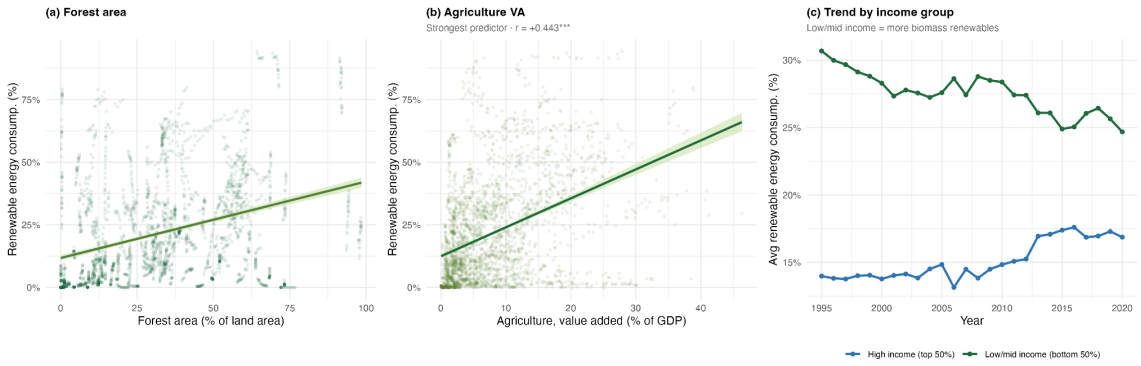


Fig. 3 — Agricultural channels: (a) forest area and (b) agriculture value added versus renewable energy consumption.

III. EMPIRICAL STRATEGY

A. Estimating Equations

I estimate the following equation as the primary specification:

$$renew_{it} = \alpha_i + \gamma_t + \beta_1 \ln g\,dp_{it} + \beta_2 (\ln g\,dp_{it})^2 + \beta_3 fdi_{it} + \beta_4 urban_{it} + \beta_5 agriva_{it} + \beta_6 co2_{it} + \beta_7 \ln p\,op_{it} + \epsilon_{it}$$

where i and t index country and year, respectively. αi are country fixed effects and γt are year fixed effects. β_1 and β_2 together test the hypothesis: a U-shaped income–renewable relationship implies $\beta_1 < 0$ and $\beta_2 > 0$, with the turning point located at GDP* = $exp[-\beta_1/(2\beta_2)]$. I report standard errors robust to arbitrary heteroskedasticity (HC1) for all cross-sectional and FE specifications.

For the first-difference estimator, I take annual differences of all time-varying regressors:

$$\Delta renew_{it} = \delta + \beta_1 \Delta \ln g\,dp_{it} + \beta_2 \Delta (\ln g\,dp_{it})^2 + \beta_3 \Delta fdi_{it} + \beta_4 \Delta urban_{it} + \beta_5 \Delta agriva_{it} + \beta_6 \Delta co2_{it} + \beta_7 \Delta \ln p\,op_{it} + \Delta \varepsilon_{it}$$

The FD estimator removes all time-invariant country characteristics without requiring explicit fixed-effect dummies. It identifies coefficients from year-to-year changes within countries. Residuals in the differenced equation may exhibit serial correlation; I therefore report Newey-West standard errors with a bandwidth of two lags.

B. Model Variants

Five specifications are reported in Table 2. Model (1) is pooled OLS with country-clustered standard errors, providing a benchmark that captures both between-country and within-country variation. Model (2) is the two-way FE estimator. Model (3) is the FD estimator. Models (4) and (5) apply the two-way FE specification separately to the subsamples of developed and developing countries, respectively, to test for heterogeneous effects.

Time-invariant or near-constant regressors (forest area, agricultural land share, temperature, precipitation) are excluded from the within-estimators because first-differencing or demeaning renders their variation negligible, inflating standard errors without adding identification. They are retained in the OLS benchmark.

C. Identification and Limitations

The two-way FE and FD estimators eliminate bias from unobserved country-level fixed characteristics (e.g., geography, historical energy endowments, institutional quality) and from global year-specific shocks. However, time-varying unobserved confounders: such as domestic energy prices, subsidy regimes, or regulatory changes, remain potential sources of bias. Moreover, reverse causality is possible: countries prioritizing renewable energy may simultaneously reduce CO_2 emissions, confounding the CO_2 coefficient. The estimates should therefore be interpreted as associations that are robust to time-invariant confounding, rather than causal effects.

IV. RESULTS

A. Overview

Table 2 reports the main regression results. The R-squared declines from 0.454 in the OLS specification to 0.347 in the two-way FE and 0.054 in the FD model. This pattern is expected: because approximately 95 percent of the variation in renew is between-country, absorbing country fixed effects removes most of the explainable variation. The low within-R-squared in Models (2) and (3) does not indicate model misspecification; the F-statistics remain highly significant ($p < 0.001$) in all specifications.

Table 2. Renewable Energy Consumption: Main Regression Results

	(1)	(2)	(3)	(4)	(5)
	OLS	**Two-Way FE**	**FD**	**FE (Dev.)**	**FE (Dev'ing)**
Economic variables					
ln GDP per capita	-99.405***	-66.747***	-20.837***	-31.873***	-82.419***
	(4.566)	(3.634)	(5.386)	(10.524)	(5.513)
ln GDP per capita²	5.436***	3.674***	1.110***	1.554***	4.914***
	(0.237)	(0.211)	(0.311)	(0.517)	(0.336)
FDI (% GDP)	-0.084***	-0.005	-0.002	-0.003	-0.042**
	(0.020)	(0.005)	(0.002)	(0.003)	(0.020)
Urban population (%)	0.074***	-0.213***	-0.404***	0.039	-0.208***
	(0.025)	(0.032)	(0.096)	(0.045)	(0.042)
Agricultural and climate variables					
Forest area (% land)	0.296***	-	-	-	-
	(0.026)	-	-	-	-
Agriculture land (% land)	-0.050***	-	-	-	-
	(0.018)	-	-	-	-
Agriculture VA (% GDP)	0.187**	0.097**	0.122***	-1.228***	0.080*
	(0.085)	(0.041)	(0.031)	(0.184)	(0.047)
Temperature (avg. °C)	0.124**	-	-	-	-
	(0.052)	-	-	-	-

Precipitation (mm/yr)	-0.001*	-	-	-	-
	(0.001)	-	-	-	-
Controls					
CO_2 per capita	-1.739***	-1.112***	-0.428***	-0.719***	-1.797***
	(0.090)	(0.067)	(0.051)	(0.051)	(0.164)
ln Population	-0.487***	-10.457***	-8.917***	-10.038***	-9.665***
	(0.173)	(0.858)	(2.266)	(0.721)	(1.491)
R-squared	0.454	0.347	0.054	0.302	0.272
Observations	3,083	3,083	2,961	1,035	2,048
Country FE	No	Yes	—	Yes	Yes
Year FE	No	Yes	—	Yes	Yes
SE type	HC1	HC1	HC1	HC1	HC1

Notes: Dependent variable: renew. Models (1)–(3) use the full sample; Models (4) and (5) restrict to high-income and low/middle-income countries, respectively. All specifications use HC1 heteroskedasticity-robust standard errors; Country and year fixed effects absorbed where indicated. Dash (—) indicates the variable is excluded because it is near-constant within-country. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B. Non-linear Income Effects

The non-linear hypothesis is supported across all five specifications. In every model, the coefficient on ln GDP per capita is negative and the coefficient on its square is positive, both significant at conventional levels across all five specifications. An F-test of joint significance rejects the null hypothesis of no nonlinearity in all cases ($p < 0.001$ in the two-way FE). The implied turning point from Model (2) is $\exp[66.75/(2 \times 3.67)] \approx \$8{,}860$ per capita; from the FD Model (3), the implied threshold is approximately \$12,000 per capita. The higher FD estimate suggests that the short-run income threshold for renewable expansion is greater than the long-run structural threshold. At the median sample income level, the marginal effect of GDP growth on renewables remains negative, indicating that most countries in the sample are still in the fossil-expansion phase of development.

C. Fossil-Fuel Lock-In

Carbon dioxide emissions per capita emerge as the most robust predictor of lower renewable energy consumption. The association is negative and statistically significant at the one-percent level in all four models. Moving from the pooled OLS ($\hat{\beta} = -1.74$) to the two-way FE (($\hat{\beta} = -1.11$) attenuates the magnitude, reflecting the fact that part of the cross-sectional correlation captures persistent differences in energy infrastructure rather than dynamic adjustment. The FD estimate (($\hat{\beta} = -0.43$) captures the year-to-year component: in a given country, a one-metric-ton increase in per-capita CO_2 emissions in year t is associated with a 0.43 pp decline in the renewable share in that same year.

D. The Agricultural Channel

The coefficient on agriculture value added is the central finding of this paper. In the OLS benchmark (Model 1), a one percentage-point increase in agriculture's share of GDP is associated

with a 0.187 pp increase in the renewable energy share ($p < 0.05$). This estimate conflates within-country dynamics with cross-sectional differences between structurally agricultural and non-agricultural economies.

Conditioning on country and year fixed effects (Model 2), the coefficient remains positive and significant ($\beta = +0.097$, $p < 0.05$), indicating that within-country agricultural growth is associated with renewable energy adoption beyond that explained by shared country-level characteristics. In the first-difference model (Model 3), the association strengthens: a one pp increases in agri_VA growth in year t is associated with a 0.122 pp increase in the renewable share in that same year ($p < 0.001$). This within-country, year-over-year estimate is the most credible from an identification standpoint, as it removes all time-invariant confounders. The result is consistent with a biomass mechanism: growth in agricultural activity generates organic residues (crop waste, livestock manure, fuelwood) that are converted into biomass energy, increasing the renewable share.

E. Urbanization: A Sign Reversal

The urban population share exhibits a well-known sign reversal between OLS and within-country estimators. In Model (1), urbanization is positively associated with renewable consumption (+0.074***, $p < 0.001$), because high-income countries tend to be simultaneously more urbanized and more invested in modern renewables (solar, wind). Once country fixed effects are absorbed, however, the sign reverses: within a given country, a one pp increase in the urban share is associated with a 0.213 pp decline in the renewable share (Model 2) and a 0.404 pp decline in the first-difference model (Model 3). The mechanism is straightforward: urbanization accompanies industrialization, which expands demand for electricity in ways that fossil fuels meet faster than renewables in the short run. The FDI coefficient undergoes a similar reversal from OLS to within-estimators, suggesting that the negative OLS coefficient reflects cross-sectional compositional differences rather than a causal effect of foreign investment.

V. HETEROGENEITY AND ROBUSTNESS

A. Development-Status Heterogeneity

Models (4) and (5) reveal substantial heterogeneity in the agricultural channel. In developing countries, the agriculture value added coefficient is positive ($\beta = +0.080$, $p = 0.047$), consistent with biomass-driven renewable adoption. In developed countries, the coefficient reverses sharply to -1.228 ($p < 0.001$), reflecting the fossil-fuel intensity of mechanized agriculture. An interaction model confirms this difference is statistically significant ($\chi^2 = 22.32$, $p < 0.001$).

B. Robustness Checks

First, replacing the dependent variable with ln(renew) leaves all core coefficients unchanged in sign and significance. Second, applying HC1 heteroskedasticity-robust standard errors across all specifications does not alter the main conclusions.

VI. CONCLUSION

This paper presents evidence on the economic and agricultural determinants of renewable energy consumption using a global panel of 122 countries over 1995–2020. Three main findings emerge. First, the relationship between income and renewable energy consumption is nonlinear and U-shaped, with a turning point of approximately $8,860–$12,000 per capita. Most developing

countries remain in the downward segment of the curve, where economic growth is still associated with fossil-fuel expansion. Second, fossil-fuel intensity proxied by CO_2 per capita is the most consistent structural obstacle to renewable transitions, operating through both persistent infrastructure differences and short-run dynamics. Third, and most novel, growth in agriculture's share of the economy is positively associated with renewable energy consumption within developing countries, consistent with a biomass energy channel, but negatively associated in developed countries where agriculture is fossil-fuel-intensive.

The estimates should be interpreted with caution. The FE and FD estimators eliminate time-invariant confounding but cannot address time-varying unobservable or reverse causality. A natural extension of this work would exploit plausibly exogenous variation, such as climate shocks as instruments for agricultural output to strengthen the causal interpretation of the agricultural channel.

APPENDIX

Figure A.1. Distribution and Time Trend of Renewable Energy Consumption

Notes: Panel (a) shows the distribution of the dependent variable (*renew*) across all country-year observations. Panel (b) plots the log-transformed outcome. Panel (c) compares the distribution across five-year intervals by income group (developed vs. developing). Panel (d) shows the global mean and median trend over 1995–2020, with the shaded band representing the interquartile range. N = 3,083 observations, 122 countries.

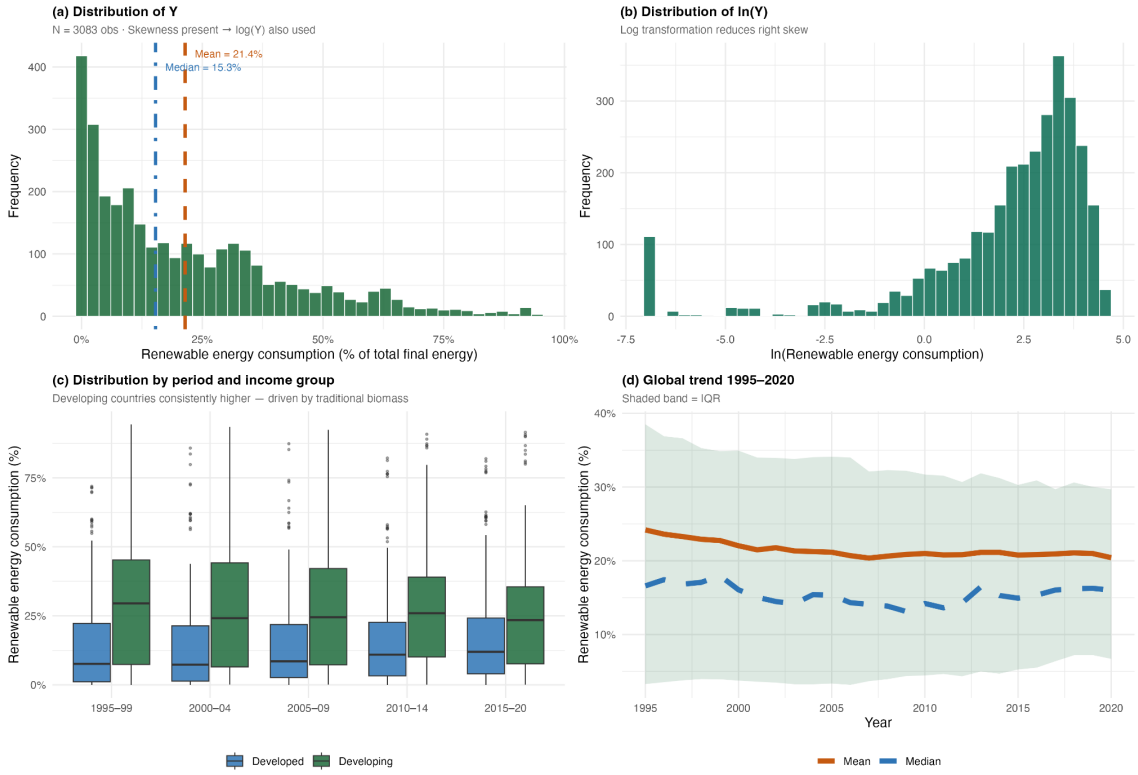


Fig. A.1 — Distribution and time trend of renewable energy consumption (%).

Figure A.2. Renewable Energy Consumption versus Economic Development

Notes: Scatter plot of renewable energy consumption (% of total final energy) against ln(GDP per capita, constant 2015 USD). The orange curve is a quadratic fit; the gray dashed line is a linear fit. N = 3,083 observations, 122 countries, 1995–2020. Sources: World Bank WDI, Our World in Data.

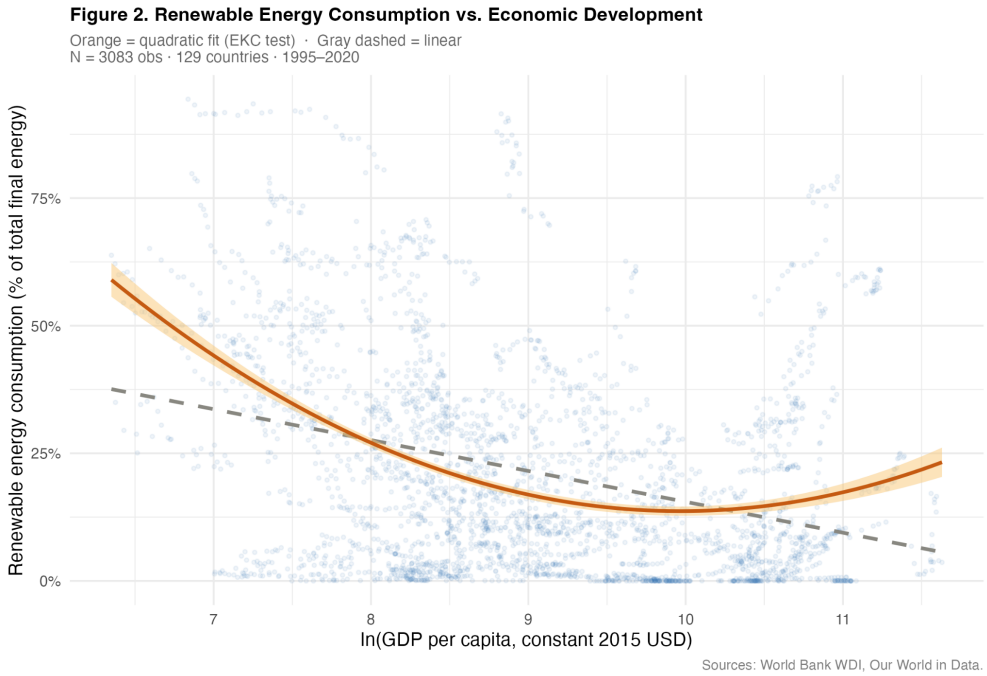


Fig. A.2 — Renewable energy consumption versus ln(GDP per capita): quadratic fit (solid) and linear fit (dashed).

Figure A.3. Agricultural Channels: Bivariate Evidence

Notes: Panel (a) plots renewable energy consumption against forest area (% of land area). Panel (b) plots against agriculture value added (% of GDP); the Pearson correlation is r = +0.443 ($p < 0.001$). Panel (c) shows average renewable energy consumption by income group over 1995–2020. All observations are country-year pairs. Sources: World Bank WDI, Our World in Data.

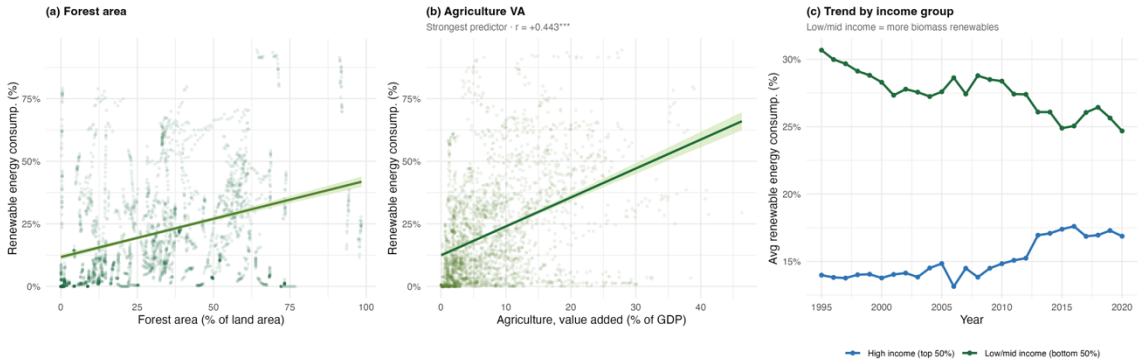


Figure A.4. Correlation Matrix (Pearson and Spearman)

Notes: Pairwise correlations among all variables used in the analysis. The left panel reports Pearson r; the right panel reports Spearman ρ. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Color intensity reflects the magnitude of the correlation.

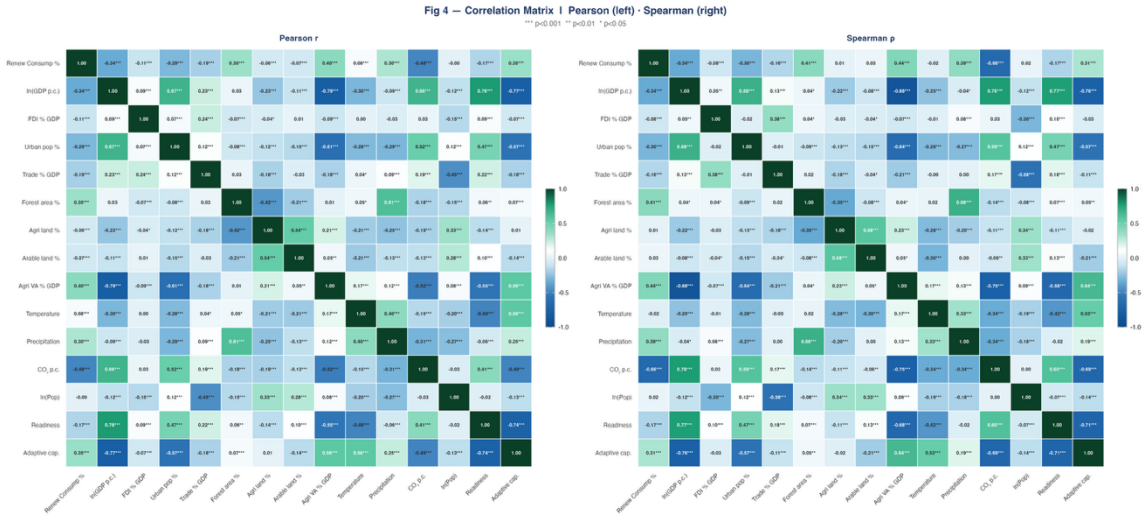
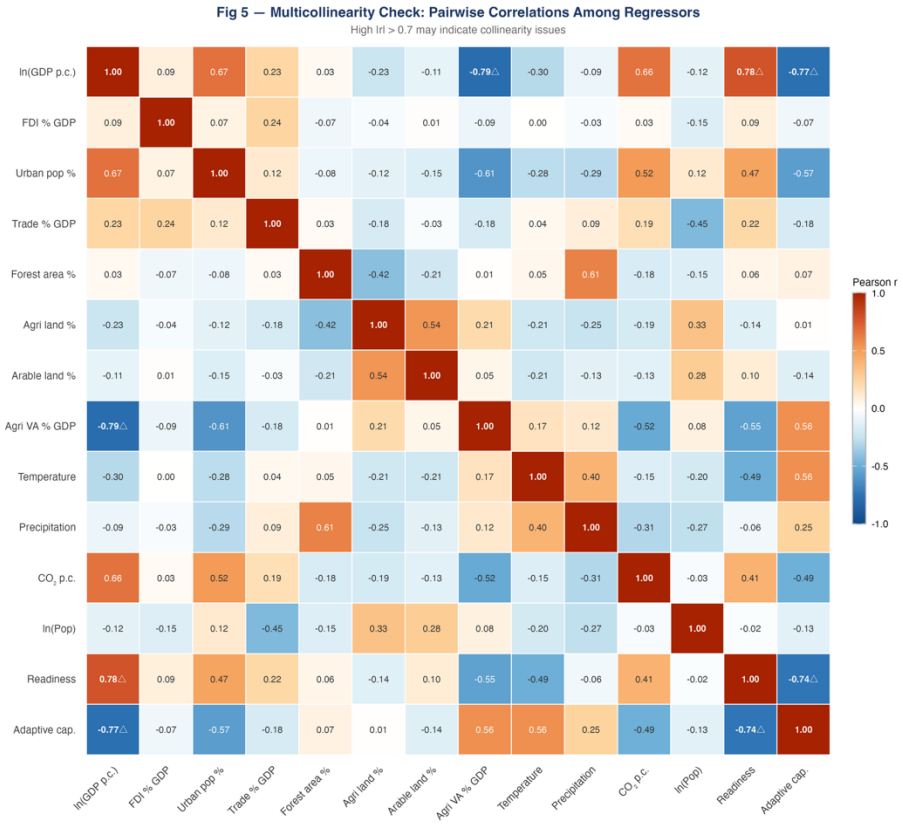


Figure A.5. Multicollinearity Check: Pairwise Correlations Among Regressors

Notes: Pearson correlations among all independent variables included in the regression specifications. Cells marked with △ indicate pairs with |r| > 0.70, flagging potential collinearity concerns. The three high-correlation pairs identified are: ln(GDP p.c.) × Agri VA (r = −0.79), ln(GDP p.c.) × Readiness (r = +0.78), and ln(GDP p.c.) × Adaptive capacity (r = −0.77).



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